



CS702: Advanced Algorithms Analysis and Design

Assignment 4

Fall 2008

Maximum Points: 100

Due Date: Tuesday, 3rd Feb. 2009

Time: 6:00 PM

Instructions

The purpose of this assignment is to give you hands on practice. It is expected that students will solve the assignments themselves. Following rules will apply during the evaluation of assignment.

- ❖ Cheating from any source will result in zero marks in the assignment.
- ❖ Any student found cheating in any of the two assignments submitted will be awarded "F" grade in the course.
- ❖ No assignment after due date will be accepted
- ❖ Full credit will be given only to correct solutions, attempt will also be appreciated.

Reading Required: Chapters 25, 31 and 32 covered from Recommended Book and Lectures 36 to 42 delivered by Virtual University of Pakistan.

Exercises: 20 × 5 = 100 Points

1. Show how to express the single-source shortest-paths problem as a product of matrices and a vector. Describe how evaluating this product corresponds to a Bellman-Ford-like algorithm (see Section 24.1 [The Bellman-Ford Algorithm]). [Exercise 25.1-5]
2. As it appears above, the Floyd-Warshall algorithm requires $\Theta(n^3)$ space, since we compute for $i, j, k = 1, 2, \dots, n$. Show that the following procedure, which simply drops all the superscripts, is correct, and thus only $\Theta(n^2)$ space is required.

FLOYD-WARSHALL' (W)

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1  $n \leftarrow \text{rows}[W]$ 
2  $D \leftarrow W$ 
3 for  $k \leftarrow 1$  to  $n$ 
4   do for  $i \leftarrow 1$  to  $n$ 
5     do for  $j \leftarrow 1$  to  $n$ 
6       do  $d_{ij} \leftarrow \min(d_{ij}, d_{ik} + d_{kj})$ 
7 return  $D$ 
[Exercise 25.2-4]
```

3. Professor Greenstreet claims that there is a simpler way to reweight edges than the method used in Johnson's algorithm. Letting $w^* = \min_{(u, v) \in E} \{w(u, v)\}$, just define $\hat{w}(u, v) = w(u, v) - w^*$ for all edges $(u, v) \in E$. What is wrong with the professor's method of reweighting? [Exercise 25.3-4]
4. Professor Michener claims that there is no need to create a new source vertex in line 1 of JOHNSON. He claims that instead we can just use $G' = G$ and let s be any vertex in $V[G]$. Give an example of a weighted, directed graph G for which incorporating the professor's

idea into JOHNSON causes incorrect answers. Then show that if G is strongly connected (every vertex is reachable from every other vertex), the results returned by JOHNSON with the professor's modification are correct. [Exercise 25.3-6]

5.

i) Prove that for all integers a , k , and n :

$$\gcd(a, n) = \gcd(a + kn, n)$$

ii) Working modulo $q = 11$, how many spurious hits does the Rabin-Karp matcher encounter in the text $T = 3141592653589793$ when looking for the pattern $p = 26$?